Exercise Set 5

MA2071

(1) Find a potential function for the vector field $F(x, y) = x^3y^4\,\mathbb{i} + x^4y^3\,\mathbb{j}$ and use it to evaluate the line integral of F along the curve $C(t) = \sqrt{t}\,\mathbb{i} + (1 + t^3)\,\mathbb{j}$ for $t \in [0, 1]$.

(2) Find a potential function for the vector field $F(x, y, z) = yz\,\mathbb{i} + xz\,\mathbb{j} + (xy + 2z)\,\mathbb{k}$ and use it to evaluate the line integral of F along the line segment from $(1, 0, -2)$ to $(4, 6, 3)$.

(3) Using Green's theorem, evaluate

$$\int_C (3y - e^{\sin x})\,dx + (7x + \sqrt{y^4 + 1})\,dy$$

where C is the circle $x^2 + y^2 = 1$.

(4) Using Green's theorem, evaluate

$$\int_C (y + e^{\sqrt{x}})\,dx + (2x + \cos y^2)\,dy$$

where C is the positively oriented boundary of the region enclosed by the parabolas $y = x^2$ and $x = y^2$.

(5) Find (a) the curl, and (b) the divergence of the following two vector fields:

(i) $F(x, y, z) = xyz\,\mathbb{i} - x^2y\,\mathbb{k}$
(ii) $F(x, y, z) = e^x \sin y\,\mathbb{i} + e^x \cos y\,\mathbb{j} + z\,\mathbb{k}$

(6) Prove that for a function $f : \mathbb{R}^3 \to \mathbb{R}$ we have $\operatorname{curl}(\nabla f) = 0$.

(7) Show that for a vector field $F : \mathbb{R}^3 \to \mathbb{R}^3$ we have $\operatorname{div}(\operatorname{curl}(F)) = 0$.

(8) Determine whether the vector field $F(x, y, z) = 2xy\,\mathbb{i} + (x^2 + 2yz)\,\mathbb{j} + y^2\,\mathbb{k}$ is conservative and, if so, find a potential function.

(9) Evaluate the surface integral $\int\int_S y\,dS$ where S is the surface $z = x + y^2$ above the rectangle $[0, 1] \times [0, 2]$.

(10) Evaluate the surface integral $\int\int_S x^2yz\,dS$ where S is part of the plane $z = 1 + 2x + 3y$ that lies above the rectangle $[0, 3] \times [0, 2]$.

(11) Compute the flux of the vector field $F(x, y, z) = xy\,\mathbb{i} + yz\,\mathbb{j} + zx\,\mathbb{k}$ across S, where S is the part of the paraboloid $z = 4 - x^2 - y^2$ that lies above the square $[0, 1] \times [0, 1]$, and has upward orientation.

(12) Evaluate the line integral $\int_C F ds$ where $F(x, y, z) = (x + y^2)\,\mathbb{i} + (y + z^2)\,\mathbb{j} + (z + x^2)\,\mathbb{k}$ and C is the triangle with vertices $\mathbb{i}$, $\mathbb{j}$ and $\mathbb{k}$.

(13) Evaluate $\int\int_S \operatorname{curl} F \cdot N\,dS$ where $F(x, y, z) = yz\,\mathbb{i} + xz\,\mathbb{j} + xy\,\mathbb{k}$ and S is the part of the paraboloid $z = 9 - x^2 - y^2$ that lies above the plane $z = 5$, oriented upwards.

(14) Calculate the flux of the vector field $F(x,\, y,\, z) = 3xy^2\,\mathbb{i} + xe^z\,\mathbb{j} + z^3\,\mathbb{k}$ across the surface S, where S is the surface of the solid bounded by the cylinder $y^2 + z^2 = 1$ and the planes $x = -1$ and $x = 2$.

(15) Calculate the flux of the vector field $F(x,\, y,\, z) = (\cos z + xy^2)\,\mathbb{i} + xe^{-z}\,\mathbb{j} + (\sin y + x^2 z)\,\mathbb{k}$ across the surface S, where S is the surface of the solid bounded by the paraboloid $z = x^2 + y^2$ and the plane $z = 4$.